

Bounds for Regret-Matching Algorithms

Amy Greenwald

*Department of Computer Science
Brown University, Box 1910
Providence, RI 02912*

AMY@BROWN.EDU

Zheng Li

*Division of Applied Mathematics
Brown University, Box F
Providence, RI 02912*

ZHENG@DAM.BROWN.EDU

Casey Marks

*Department of Computer Science
Brown University, Box 1910
Providence, RI 02912*

CASEY@CS.BROWN.EDU

Abstract

We study a general class of learning algorithms, which we call regret-matching algorithms, along with a general framework for analyzing their performance in online (sequential) decision problems (ODPs). In each round of an ODP, an agent chooses a probabilistic action and receives a reward. The particular reward function that applies at any given round is not revealed until after the agent acts. Further, the reward function may change arbitrarily from round to round.

Our analytical framework is based on a set Φ of transformations over the agent's set of actions. We calculate a Φ -regret vector by comparing the reward obtained by an agent over some finite sequence of rounds to the reward that could have been obtained had the agent instead played each transformation $\phi \in \Phi$ of its sequence of actions. Regret-matching algorithms select the agent's next action based on the vector of Φ -regrets together with a link function f .

In this paper, we derive bounds on the regret experienced by (Φ, f) -regret matching algorithms for polynomial and exponential link functions and arbitrary Φ . Whereas others have typically derived bounds on *distribution* regret, our focus is on bounding the expectation of *action* regret. A simplification of our framework, however, yields bounds on distribution regret equivalent to (and in some cases slightly stronger than) others that have appeared in the literature.

1. Introduction

We present a general class of learning algorithms along with a general framework for analyzing their performance in online (sequential) decision problems (ODPs). In each round of an ODP, an agent chooses a mixed strategy (i.e., a distribution over its actions), samples and plays a (pure) action, and receives a reward corresponding to its play. A reward function dictates the relationship between rewards and actions, but the particular reward function that applies at any given round is not revealed until after the agent acts. Further, the reward function may change arbitrarily from round to round; no underlying assumptions govern the reward dynamics.

Online decision problems encompass a wide variety of machine learning settings. Consider an agent learning in an infinitely repeated matrix game, for example. The reward dynamics are jointly determined by the matrix and the behavior of the other players. An ODP—with arbitrary reward dynamics—is sufficiently general to model the nonstationarity of multiagent learning.

We study regret-matching algorithms, a class of “regret-minimization” algorithms. The power of regret-matching algorithms is that they minimize regret in any ODP, that is, an agent learning via regret-matching in an arbitrary ODP does not experience too much regret for not having played alternative sequences of actions other than the one it did in fact play.

Following Greenwald et al. (2006a), we adopt an analytical framework based on a set Φ of transformations over the agent’s set of actions. We calculate a Φ -regret vector by comparing the cumulative reward obtained by an agent over some finite sequence of rounds to the cumulative reward that could have been obtained had the agent instead played each transformation $\phi \in \Phi$ of its sequence of actions. Regret-matching algorithms select the agent’s next action based on the vector of Φ -regrets and a link function f . Many well-studied learning algorithms can be viewed as instances of regret matching (e.g., Freund and Schapire (1997) and Foster and Vohra (1999)).

Like Cesa-Bianchi and Lugosi (2003), we derive bounds on the regret experienced by regret matching for polynomial and exponential link functions. However, whereas they derive bounds on the *distribution* regret (i.e., the expected regret, given the agent’s mixed strategy), we bound the expectation of the *action* regret (i.e., the actual regret of the agent’s realized action). While neither style of bound is stronger than the other, a simplification of our framework yields bounds equivalent to (and in some cases slightly stronger than) theirs. We also derive the optimal parameter settings for both polynomial and exponential regret-matching algorithms.

Our approach is unusual in that we bound action regret. Most studies of action regret appear in the game-theoretic literature and achieve convergence results. Most studies of distribution regret appear in the computer science literature and achieve bounding results.

Unlike Cesa-Bianchi and Lugosi (2003), our analysis does not rely on Taylor’s theorem. Hence, we can naturally analyze algorithms based on a wider variety of link functions, in particular non-differentiable link functions, such as the class of polynomial link functions.

This paper is organized as follows. In Section 2, we formally present our analytical framework, building up to the definition of regret-matching algorithms, which is presented in Section 3. In Section 3, we also prove the regret-matching theorem, which states that regret-matching algorithms satisfy a property related to Blackwell’s condition for approachability (1956), applied in the regret domain. In Section 4, we present our main analytical tool, which we apply to derive and optimize bounds for specific regret-matching algorithms, namely those based on polynomial and exponential link functions. Section 4.3 is of particular interest, because there we show how our framework can be simplified to derive the results in Cesa-Bianchi and Lugosi (2003). In Section 5, we discuss the wide variety of related work. Proofs of select theorems appear in Appendix A.

2. Formal Framework

In this section, we present formal definitions of the key concepts in our analytical framework, namely: online decision problems, action transformations, action and distribution regret, and link functions.

2.1 Online Decision Problems

An online decision problem is parameterized by a *reward system* $(A, \mathcal{R})$, where A is a set of actions and $\mathcal{R}$ is a set of rewards. A *real-valued* reward system is one in which $\mathcal{R} \subseteq \mathbb{R}$. When we restrict our attention to *bounded* (real-valued) reward systems, we assume WLOG that $\mathcal{R} = [0, 1]$. Given a reward system $(A, \mathcal{R})$, we let $\Pi \equiv \mathcal{R}^A$ denote the set of *reward functions*.

In this paper, we assume that the agent’s action set A is finite. We denote by $\Delta(A)$ the set of probability distributions over the set A , and we allow agents to play *mixed strategies*, which means that rather than selecting an action $a \in A$ to play at each round, the agent learns a probability distribution $q \in \Delta(A)$. Hence, round t (for $t = 1, 2, \dots$) proceeds as follows:

1. the agent chooses a mixed strategy $q_t \in \Delta(A)$,

2. the agent plays an (pure) action $a_t \sim q_t$,
3. the agent receives reward $r_t(a_t) \in \mathcal{R}$,
4. the agent is informed of $r_t \in \Pi$.

The last step, in which the agent is informed of the reward function r_t (i.e., the rewards associated with actions it did not choose to play) characterizes *informed* ODPs, which are the subject of this work. Without this step, informing the agent of only $r_t(a_t)$, instead the setup would be that of a *naïve* ODP. See Auer et al. (1995), for example, for consideration of the naïve setting.

Definition 1 *Given a reward system $(A, \mathcal{R})$, a particular instance of an online decision problem can be described by a reward schedule, that is, a sequence of reward functions $\{r_t\}_{t=1}^\infty$, where $r_t \in \Pi$, so that $r_t(a) \in \mathcal{R}$ corresponds to the reward for playing action a at time $t \geq 1$. A **bounded ODP** is an ODP over a bounded reward system.*

Given a reward system $(A, \mathcal{R})$, the set of histories of length $t \geq 1$ is given by $H_t \equiv A^t \times \Pi^t$. We define H_0 to be a singleton and we let $\mathcal{H} = \cup_{t=0}^\infty H_t$ denote the complete set of histories.

Given an ODP, the particular history $(\{a_\tau\}_{\tau=1}^t, \{r_\tau\}_{\tau=1}^t)$ corresponds to the agent playing a_τ and observing reward function r_τ at each time $\tau = 1, \dots, t$. An *online learning algorithm* is a sequence of functions $\{L_t\}_{t=1}^\infty$, where $L_t : H_{t-1} \rightarrow \Delta(A)$ so that $L_t(h) \in \Delta(A)$ represents the agent's mixed strategy at time $t \geq 1$, having observed history $h \in H_{t-1}$.

2.2 Action Transformations

Fix a (finite) set of actions A . An *action transformation* is defined as a function $\phi : A \rightarrow \Delta(A)$. We let $\Phi_{\text{ALL}} \equiv \Phi_{\text{ALL}}(A)$ denote the set of all action transformations over the set A . Following Blum and Mansour (2005), we also let $\Phi_{\text{SWAP}} \equiv \Phi_{\text{SWAP}}(A) \subseteq \Phi_{\text{ALL}}$ denote the set of all action transformations that map actions to distributions with all their weight on a single action: i.e., pure strategies.

There are two well-studied subsets of Φ_{SWAP} : the set of external transformations and the set of internal transformations. Let $\delta_a \in \Delta(A)$ denote the distribution with all its weight on action a . An *external* action transformation is simply a constant transformation, so for $a \in A$,

$$\phi_{\text{EXT}}^{(a)} : x \mapsto \delta_a, \quad \text{for all } x \in A \quad (1)$$

An *internal* action transformation behaves like the identity, except on one particular input, so for $a, b \in A$

$$\phi_{\text{INT}}^{(a,b)} : x \mapsto \begin{cases} \delta_b & \text{if } x = a \\ \delta_x & \text{otherwise} \end{cases} \quad (2)$$

We let $\Phi_{\text{EXT}} \equiv \Phi_{\text{EXT}}(A)$ denote the set of external transformations and $\Phi_{\text{INT}} \equiv \Phi_{\text{INT}}(A)$ denote the set of internal transformations. Observe that $|\Phi_{\text{SWAP}}| = |A|^{|A|}$, $|\Phi_{\text{INT}}| = |A|^2 - |A| + 1$, and $|\Phi_{\text{EXT}}| = |A|$.

An action transformation can be extended so that its domain is the set of mixed strategies. Given an action transformation $\phi : A \rightarrow \Delta(A)$, let $[\phi] : \Delta(A) \rightarrow \Delta(A)$ be the linear transformation given by

$$[\phi](q) = \sum_{a \in A} q(a) \phi(a) \quad (3)$$

for all $q \in \Delta(A)$. Since $[\phi]$ is a finite-dimensional linear transformation, we may think of it as an $|A| \times |A|$ stochastic matrix; similarly, we may think of $\phi(a)$ as a real-valued vector of length $|A|$.

2.3 The Regret Vector

A regret vector is calculated with respect to a set of action transformations. Each entry is calculated as the difference between the rewards associated with the agent’s “play” and a transformation of that “play.” Taking the play to be the actual action played, a_t , yields *action* regret, whereas taking the play to be the mixed strategy q_t yields *distribution* regret.

More precisely, given a reward system $(A, \mathcal{R})$, a reward function $r \in \Pi$, an action transformation $\phi \in \Phi_{\text{ALL}}$, and an action $a \in A$, the (action) ϕ -regret is given by $\rho^\phi(a, r) = r(\phi(a)) - r(a)$, where $r(q) = \mathbb{E}_{a \sim q} [r(a)]$, for all $q \in \Delta(A)$. This quantity is the difference between the rewards the agent obtains by playing action a and the expected rewards that the agent could have obtained by playing the transformed action $\phi(a)$. Given a set of action transformations $\Phi \subseteq \Phi_{\text{ALL}}$, the (action) Φ -regret vector is given by $\rho^\Phi(a, r) = (\rho^\phi(a, r))_{\phi \in \Phi}$. In this work, we restrict our attention to finite action transformation sets $\Phi \subseteq \Phi_{\text{ALL}}$ so that, for real-valued reward systems, the Φ -regret vector is an element of finite-dimensional Euclidean space.¹

Given an ODP $\{r_\tau\}_{\tau=1}^\infty$ and an infinite sequence of the agent’s actions $\{a_\tau\}_{\tau=1}^\infty$, the (action) cumulative Φ -regret vector at time $t \geq 1$, where $\Phi \subseteq \Phi_{\text{ALL}}$, is computed as follows:

$$R_t^\Phi(\{a_\tau\}, \{r_\tau\}) = \sum_{\tau=1}^t \rho^\Phi(a_\tau, r_\tau) \quad (4)$$

The cumulative Φ -regret vector compares the cumulative rewards obtained by the agent with the rewards that the agent could have expected to obtain by consistently transforming each of its actions according to each $\phi \in \Phi$. We define $R_0^\Phi(\{a_\tau\}, \{r_\tau\}) = 0$.

We sometimes treat cumulative regret as a function from histories of length T to regret vectors, in which case we write $R_t^\Phi(h)$, for $h \in H_T$ and $t \leq T$. When considering a particular ODP $\{r_t\}_{t=1}^\infty$ together with an online learning algorithm $\{L_t\}_{t=1}^\infty$, we treat cumulative regret as a random vector over a probability space whose universe consists of infinite sequences of the agent’s actions $\{a_\tau\}_{\tau=1}^\infty$ and whose measure is defined by the learning algorithm. In this case, we write R_t^Φ and $\rho_t^\Phi \equiv \rho^\Phi(a_t, r_t)$ for the cumulative and instantaneous Φ -regret vectors, respectively. The quantity we seek to bound is the expected value of time-averaged (action) regret at time t , that is

$$\mathbb{E} \left[\frac{1}{t} \max_{\phi \in \Phi} R_t^\phi \right] \quad (5)$$

The distribution regret at time t is calculated with respect to the mixed strategy q_t that the agent learns, rather than the action a_t that it actually plays. We denote the instantaneous and cumulative distribution Φ -regret vectors by $\hat{\rho}_t^\Phi$ and $\hat{R}_t^\Phi$, respectively. In particular, for $\phi \in \Phi$,

$$\hat{\rho}^\phi(q, r) = \mathbb{E}_{a \sim q} [\rho^\phi(a, r)] \quad (6)$$

$$\hat{\rho}_t^\phi = \hat{\rho}^\phi(q_t, r_t) \quad (7)$$

$$\hat{R}_t^\phi = \sum_{\tau=1}^t \hat{\rho}_\tau^\phi \quad (8)$$

The quantity bounded by Cesa-Bianchi and Lugosi (2003) is the time-averaged distribution regret, namely

$$\frac{1}{t} \max_{\phi \in \Phi} \hat{R}_t^\phi \quad (9)$$

1. Any element of Φ_{ALL} can be expressed as a convex combination of elements of Φ_{SWAP} ; hence, for bounding results, it suffices to study Φ_{SWAP} . For convergence results, it suffices to study Φ_{INT} (Greenwald et al., 2006a).

2.4 Link Functions

We denote the positive and negative orthants of $\mathbb{R}^n$ by $\mathbb{R}_+^n \equiv \{x \in \mathbb{R}^n \mid x_i \geq 0, \forall i\}$ and $\mathbb{R}_-^n \equiv \{x \in \mathbb{R}^n \mid x_i \leq 0, \forall i\}$, respectively. A *link* function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a function that is subgradient to a convex (*potential*) function $F : \mathbb{R}^n \rightarrow \mathbb{R}$. It can be shown that the co-domain of any link function f is the positive orthant $\mathbb{R}_+^n$ if f is subgradient to a convex function $F : \mathbb{R}^n \rightarrow \mathbb{R}$ that is bounded on the negative orthant: i.e., $F(\mathbb{R}_-^n) \subseteq (-\infty, k]$, for some $k < \infty$.

Although our main analytical tool, which bounds the regret accumulated by regret-matching algorithms, can be applied more broadly, we demonstrate its efficacy on two classes of link functions, polynomial and exponential, which give rise to two well-studied classes of regret-matching algorithms. The polynomial link function is defined by $f_i(x) = (x_i^+)^{p-1}$ with parameter $p \in \mathbb{R}$. (For all $a \in \mathbb{R}$, $a^+ = \max\{a, 0\}$.) The exponential link function is defined by $f_i(x) = e^{\eta x_i}$ with parameter $\eta \in \mathbb{R}$.

3. Regret-Matching Algorithms

In this section, we define a general class of online learning algorithms, which we call regret-matching algorithms,² that are parameterized by a set of action transformations Φ and a link function f . In addition, we prove the regret-matching theorem, which states that the algorithms in this class satisfy the generalized Blackwell regret condition, also parameterized by Φ and f .

3.1 Generalized Blackwell Regret Condition

First, we define the generalized Blackwell regret condition.

Given a repeated vector-valued matrix game with rewards in $\mathbb{R}^n$, Blackwell’s seminal approachability theorem (1956) provides a sufficient condition for the “approachability” of a convex subset of $\mathbb{R}^n$ by each player’s time-averaged reward vector. If the vector-valued rewards are given by the Φ -regret vectors generated by playing an underlying real-valued matrix game, the approachability of the negative orthant is known as the “no- Φ -regret” property (see Definition 19). Applying Blackwell’s sufficient condition in this setting yields the “Blackwell Φ -regret condition.” Here we present the Blackwell (Φ, f) -regret condition, which, if the link function is taken to be the polynomial link function with $p = 2$, reduces to the Blackwell Φ -regret condition.

Definition 2 (Generalized Blackwell Regret Condition) *Fix a reward system $(A, \mathcal{R})$, a finite set $\Phi \subseteq \Phi_{ALL}$ of action transformations, and a link function $f : \mathbb{R}^\Phi \rightarrow \mathbb{R}_+^\Phi$. An online learning algorithm $\{L_t\}_{t=1}^\infty$ is said to satisfy the Blackwell (Φ, f) -regret condition if*

$$f(R_{t-1}^\Phi(h)) \cdot \mathbb{E}_{a \sim L_t(h)} [\rho^\Phi(a, r)] \leq 0 \quad (10)$$

for all times $t \geq 1$, for all histories $h \in H_{t-1}$ of length $t-1$, and for all reward functions $r : A \rightarrow \mathcal{R}$.

3.2 Regret-Matching Algorithms

Second, we define regret-matching algorithms.

Fix a real-valued reward system $(A, \mathcal{R})$, a finite set $\Phi \subseteq \Phi_{ALL}$ of action transformations, and a link function $f : \mathbb{R}^\Phi \rightarrow \mathbb{R}_+^\Phi$. For any history $h \in H_{t-1}$, recall that the quantity $R_{t-1}^\Phi(h)$ denotes the cumulative regret, through time $t-1$, for not having played according to the action transformations contained in Φ . If $R_{t-1}^\Phi(h) \in \mathbb{R}_+^\Phi$, so that each element of the regret vector is non-positive, then the agent does not regret its past actions. In this case, a (Φ, f) -regret-matching algorithm leaves the agent’s next play unspecified. But if $R_{t-1}^\Phi(h) \notin \mathbb{R}_+^\Phi$, so that the agent “feels” regret in at least one dimension, then applying the link function f to this quantity yields a non-negative vector,

2. We appropriate this terminology from Hart and Mas-Colell (2001), whose regret-matching algorithms based on Φ_{EXT} and the polynomial link functions are instances of this class.

Algorithm 1 (Φ, f) -RegretMatchingAlgorithm(ODP $\{r_t\}_{t=1}^\infty$, numRounds T)

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1: initialize  $X_0^\Phi = 0$ 
2: for  $t = 1, \dots, T$  do
3:   sample pure action  $a_t \sim q_t$ 
4:   observe reward function  $r_t$ 
5:   for all  $\phi \in \Phi$  do
6:     compute instantaneous regret  $x_t^\phi = \rho^\phi(a_t, r_t)$ 
7:     update cumulative regret vector  $X_t^\phi = X_{t-1}^\phi + x_t^\phi$ 
8:   end for
9:   let  $Y_t^\Phi = f(X_t^\Phi)$ 
10:  if  $Y_t^\Phi = 0$  then
11:    set  $q_{t+1} \in \Delta(A)$  arbitrarily
12:  else
13:    let  $M_t = \sum_{\phi \in \Phi} Y_t^\phi[\phi] / \sum_{\phi \in \Phi} Y_t^\phi$ 
14:    solve for a fixed point  $q_{t+1} = M_t(q_{t+1})$ 
15:  end if
16: end for

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call it $Y_t^\Phi \in \mathbb{R}_+^\Phi$. Normalizing this vector, we compute the linear transformation M_t as a convex combination of the linear transformations $[\phi]$ as follows:

$$M_t = \frac{\sum_{\phi \in \Phi} Y_t^\phi[\phi]}{\sum_{\phi \in \Phi} Y_t^\phi} \quad (11)$$

The linear transformation M_t maps $\Delta(A)$, a nonempty compact convex set, into itself. Moreover, M_t is continuous, since all $\phi \in \Phi$ are linear functions in finite-dimensional Euclidean space, and hence continuous. Therefore, by Brouwer's fixed point theorem, M_t has a fixed point $q \in \Delta(A)$. An action (Φ, f) -regret-matching algorithm plays q whenever $R_{t-1}^\Phi(h) \notin \mathbb{R}_-^\Phi$.

Definition 3 (Regret-Matching Algorithm) *An action (Φ, f) -regret-matching algorithm plays the fixed point of M_t (defined in Equation 11), whenever $R_{t-1}^\Phi(h) \notin \mathbb{R}_-^\Phi$.*

The pseudocode for the class of action (Φ, f) -regret-matching algorithms is shown in Algorithm 1. The cumulative regret vector is initialized to zero. For all times $t = 1, \dots, T$, the agent samples a pure action a_t according to the distribution q_t , after which it observes the reward function r_t . Given a_t and r_t , the agent computes its instantaneous regret with respect to each $\phi \in \Phi$, and updates the cumulative Φ -regret vector accordingly. A subroutine is then called to compute the mixed strategy that the agent learns to play at time $t + 1$. In this subroutine, the link function f is applied to the cumulative Φ -regret vector. (Recall that the co-domain of a link function is the positive orthant.) If this quantity is zero, then the subroutine returns an arbitrary mixed strategy. Otherwise, the subroutine returns a fixed point of the stochastic matrix M_t .

Many well-known online learning algorithms arise as instances of action or distribution (Φ, f) -regret matching. (Distribution regret matching is defined in Section 4.3.) The no-external-regret algorithm of Hart and Mas-Colell (2000) is the special case of (Φ, f) -action-regret matching in which $\Phi = \Phi_{\text{EXT}}$ and f is the polynomial link function with $p = 2$. The no-internal-regret algorithm of Foster and Vohra (1999) is equivalent to (Φ, f) -distribution-regret matching with $\Phi = \Phi_{\text{INT}}$ and the polynomial link function with $p = 2$. If f is the exponential link function, then (Φ, f) -distribution-regret reduces to Freund and Schapire's Hedge algorithm (1997) when $\Phi = \Phi_{\text{EXT}}$ and a variant of an algorithm discussed by Cesa-Bianchi and Lugosi (2003) when $\Phi = \Phi_{\text{INT}}$. For a more thorough comparison of these and related algorithms, see Section 5.

3.3 Regret Matching Theorem

We now prove the regret matching theorem, which states that regret-matching algorithms satisfy the generalized Blackwell regret condition (with equality).

Theorem 4 (Regret-Matching Theorem) *Given a real-valued reward system $(A, \mathcal{R})$, a finite set $\Phi \subseteq \Phi_{ALL}$ of action transformations, and a link function $f : \mathbb{R}^\Phi \rightarrow \mathbb{R}_+^\Phi$, the (Φ, f) -regret-matching algorithm $\{L_t\}_{t=1}^\infty$ satisfies the generalized (Φ, f) -Blackwell regret condition (with equality).*

Proof For all times $t \geq 1$ and for all histories $h \in H_{t-1}$, we abbreviate as follows: $Y_t^\Phi \equiv f(R_{t-1}^\Phi(h))$. Since f is a link function, Y_t^Φ is non-negative. If Y_t^Φ is the zero vector, the result follows immediately. Otherwise, for all reward functions $r : A \rightarrow \mathcal{R}$,

$$Y_t^\Phi \cdot \mathbb{E}_{a \sim L_t(h)} [\rho^\Phi(a, r)] \quad (12)$$

$$= \sum_{\phi \in \Phi} Y_t^\phi (r \cdot [\phi](L_t(h)) - r \cdot L_t(h)) \quad (13)$$

$$= r \cdot \left(\sum_{\phi \in \Phi} Y_t^\phi [\phi](L_t(h)) - \sum_{\phi \in \Phi} Y_t^\phi L_t(h) \right) \quad (14)$$

$$= r \cdot \sum_{\phi \in \Phi} Y_t^\phi (M_t(L_t(h)) - L_t(h)) \quad (15)$$

$$= 0 \quad (16)$$

Line (16) follows from the definition of a (Φ, f) -regret-matching algorithm, which ensures that $L_t(h)$ is a fixed point of M_t : i.e., $M_t(L_t(h)) - L_t(h) = 0$. $\blacksquare$

4. Bounds

In this section, as a consequence of the regret-matching theorem, we derive a general bound on the performance of regret-matching algorithms. We instantiate this general bound to derive specific bounds for particular choices of (classes of) link functions and particular sets of action transformations. Our foundational result is a corollary of (what is essentially) Gordon's Gradient Descent Theorem (2005), which bounds the growth rate of any real-valued function on $\mathbb{R}^n$.

Definition 5 A Gordon triple $\langle G, g, \gamma \rangle$ consists of three functions $G : \mathbb{R}^n \rightarrow \mathbb{R}$, $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$, and $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}^n$, $G(x + y) \leq G(x) + g(x) \cdot y + \gamma(y)$.

Given a Gordon triple $\langle G, g, \gamma \rangle$, if G is smooth and g is equal to the gradient of G , then $\gamma(y)$ is a bound on the higher order terms of the Taylor expansion of $G(x + y)$.

Theorem 6 (Gordon, 2005) *Assume $\langle G, g, \gamma \rangle$ is a Gordon triple and $C : \mathbb{N} \rightarrow \mathbb{R}$. Let $X_0 \in \mathbb{R}^n$, let $x_1, x_2, \dots$ be a sequence of random vectors over $\mathbb{R}^n$, and define $X_t = X_{t-1} + x_t$ for all times $t \geq 1$.*

If, for all times $t \geq 1$,

$$g(X_{t-1}) \cdot \mathbb{E}[x_t \mid X_{t-1}] + \mathbb{E}[\gamma(x_t) \mid X_{t-1}] \leq C(t) \quad a.s. \quad (17)$$

then, for all times $t \geq 0$,

$$\mathbb{E}[G(X_t)] \leq G(X_0) + \sum_{\tau=1}^t C(\tau) \quad (18)$$

Proof The proof is by induction on t . At time $t = 0$, $\mathbb{E}[G(X_t)] = G(X_0)$ and $\sum_{\tau=1}^t C(\tau) = 0$. At time $t \geq 1$, since $X_t = X_{t-1} + x_t$ and $\langle G, g, \gamma \rangle$ is a Gordon triple,

$$G(X_t) = G(X_{t-1} + x_t) \quad (19)$$

$$\leq G(X_{t-1}) + g(X_{t-1}) \cdot x_t + \gamma(x_t) \quad (20)$$

By Assumption 17, taking conditional expectations w.r.t. X_{t-1} yields

$$\mathbb{E}[G(X_t) \mid X_{t-1}] \leq G(X_{t-1}) + g(X_{t-1}) \cdot \mathbb{E}[x_t \mid X_{t-1}] + \mathbb{E}[\gamma(x_t) \mid X_{t-1}] \quad (21)$$

$$\leq G(X_{t-1}) + C(t) \quad \text{a.s.} \quad (22)$$

Taking expectations and applying the law of iterated expectations yields

$$\mathbb{E}[G(X_t)] = \mathbb{E}[\mathbb{E}[G(X_t) \mid X_{t-1}]] \quad (23)$$

$$\leq \mathbb{E}[G(X_{t-1})] + C(t) \quad (24)$$

Therefore, by the induction hypothesis,

$$\mathbb{E}[G(X_t)] \leq \mathbb{E}[G(X_{t-1})] + C(t) \quad (25)$$

$$\leq G(X_0) + \sum_{\tau=1}^{t-1} C(\tau) + C(t) \quad (26)$$

$$= G(X_0) + \sum_{\tau=1}^t C(\tau) \quad (27)$$

■

We apply Gordon's theorem to derive the following corollary:

Corollary 7 *Given a real-valued reward system $(A, \mathcal{R})$ and a finite set $\Phi \subseteq \Phi_{ALL}$ of action transformations. If $\langle G, g, \gamma \rangle$ is a Gordon triple, then a (Φ, g) -regret-matching algorithm $\{L_t\}_{t=1}^\infty$ guarantees*

$$\mathbb{E}[G(R_t^\Phi)] \leq G(0) + t \sup_{a \in A, r \in \Pi} \gamma(\rho^\Phi(a, r)) \quad (28)$$

at all times $t \geq 0$.

Proof For all times $t \geq 1$, for all histories $h \in H^{t-1}$, and for all reward functions $r : A \rightarrow \mathcal{R}$, $g(R_{t-1}^\Phi(h)) \cdot \mathbb{E}_{a \sim L_t(h)}[\rho^\Phi(a, r)] = 0$, by the regret-matching theorem. The only part of the history through time $t-1$ that impacts the agent's mixed strategy at time t is the cumulative regret vector; hence, we rewrite $\mathbb{E}_{a \sim L_t(h)}[\rho^\Phi(a, r)]$ as $\mathbb{E}[\rho_t^\Phi \mid R_{t-1}^\Phi]$. It follows that $g(R_{t-1}^\Phi) \cdot \mathbb{E}[\rho_t^\Phi \mid R_{t-1}^\Phi] = 0$. Now apply Theorem 6 with $x_t = \rho_t^\Phi$, $X_t = R_t^\Phi$, and $C(t) = \sup_{a, r} \gamma(\rho^\Phi(a, r))$, noting that $R_0^\Phi = 0$. ■

We now apply Corollary 7 to particular Gordon triples to derive bounds on the regret experienced by agents that employ polynomial and exponential regret-matching algorithms. In doing so, we rely on three Gordon triples. (The proofs that they are indeed Gordon triples are omitted.)

4.1 Polynomial Regret-Matching Bounds

The polynomial regret-matching algorithms are based on the polynomial link function $f_i(x) = (x_i^+)^{p-1}$ with parameter $p \in \mathbb{R}$. We divide our analyses of these algorithms into two disjoint cases: $2 < p < \infty$ and $1 \leq p \leq 2$.

Given a (finite) set of actions A and a finite set of action transformations $\Phi \subseteq \Phi_{\text{ALL}}$, the *maximal activation*, denoted $\mu(\Phi)$, is computed by maximizing, over all actions $a \in A$, the number of transformations ϕ that alter action a : i.e.,

$$\mu(\Phi) = \max_A |\{\phi \in \Phi : \phi(a) \neq \delta_a\}| \quad (29)$$

Clearly, $\mu(\Phi) \leq |\Phi|$. In addition, observe that $\mu(\Phi_{\text{EXT}}) = \mu(\Phi_{\text{INT}}) = |A| - 1$.

We are now ready to bound the Φ -regret of polynomial regret matching, that is, regret matching with the polynomial link function $f_i(x) = (x_i^+)^{p-1}$. We state two theorems, the first pertaining to $2 < p < \infty$ and the second pertaining to $1 \leq p \leq 2$.

Lemma 8 *For $2 < p < \infty$, if $G(x) = \|x^+\|_p^2$,*

$$g_i(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{2x_i^{p-1}}{\|x^+\|_p^{p-2}} & \text{otherwise} \end{cases} \quad (30)$$

and $\gamma(x) = (p-1)\|x\|_p^2$, then $\langle G, g, \gamma \rangle$ is a Gordon triple.

Proof See Greenwald et al. (2006b), Appendix B, Lemma 11.

Theorem 9 *Given a bounded ODP, a finite set of action transformations $\Phi \subseteq \Phi_{\text{ALL}}$, and a polynomial link function f with $p > 2$, a (Φ, f) -regret-matching algorithm guarantees*

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq \sqrt{\frac{p-1}{t}} \sqrt[p]{\mu(\Phi)} \quad (31)$$

at all times $t \geq 1$.

Proof See Appendix A.

Lemma 10 *For $1 \leq p \leq 2$, if $G(x) = \|x^+\|_p^p$, $g_i(x) = p(x_i^+)^{p-1}$, and $\gamma(x) = \|x\|_p^p$, then $\langle G, g, \gamma \rangle$ is a Gordon triple.*

Proof See Greenwald et al. (2006b), Appendix B, Lemma 13.

Theorem 11 *Given a bounded ODP, a finite set of action transformations $\Phi \subseteq \Phi_{\text{ALL}}$, and a polynomial link function f with $1 \leq p \leq 2$, a (Φ, f) -regret-matching algorithm guarantees*

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq t^{(\frac{1}{p}-1)} \sqrt[p]{\mu(\Phi)} \quad (32)$$

at all times $t \geq 1$.

Proof See Appendix A.

4.2 Exponential Regret-Matching Bounds

Next, we bound the Φ -regret of exponential regret-matching, that is, regret-matching with the exponential link function $f_i(x) = e^{\eta x_i}$ with parameter $\eta \in \mathbb{R}$.

Lemma 12 *If*

$$G(x) = \frac{1}{\eta} \ln \left(\sum_i e^{\eta x_i} \right) \quad (33)$$

$$g_i(x) = \frac{e^{\eta x_i}}{\sum_j e^{\eta x_j}} \quad (34)$$

and $\gamma(x) = \frac{\eta}{2} \|x\|_\infty^2$, then $\langle G, g, \gamma \rangle$ is a Gordon triple.

Proof See Greenwald et al. (2006b), Appendix B, Lemma 15.

Theorem 13 *Given a bounded ODP, a finite set of action transformations $\Phi \subseteq \Phi_{ALL}$, and an exponential link function f with parameter $\eta > 0$, a (Φ, f) -regret-matching algorithm guarantees*

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq \frac{\ln |\Phi|}{\eta t} + \frac{\eta}{2} \quad (35)$$

at all times $t \geq 1$.

Proof See Appendix A.

4.3 Distribution Regret

Here, we adapt our framework to analyze distribution rather than action regret matching. We obtain bounds on distribution regret that are analogous to our bounds on action regret. For exponential distribution regret matching, our bounds match those derived by Cesa-Bianchi and Lugosi (2003); for polynomial distribution regret matching, our bounds are slightly stronger.³

A distribution-regret-matching algorithm is identical to an action-regret-matching algorithm, but uses $\hat{R}_t^\Phi$ in place of R_t^Φ .

Definition 14 (Distribution Regret Matching Algorithm) *A (Φ, f) -distribution-regret-matching algorithm plays the fixed point of*

$$\hat{M}_t = \frac{\sum_{\phi \in \Phi} f_\phi \left(\hat{R}_{t-1}^\Phi(h) \right) [\phi]}{\sum_{\phi \in \Phi} f_\phi \left(\hat{R}_{t-1}^\Phi(h) \right)} \quad (36)$$

whenever $\hat{R}_{t-1}^\Phi(h) \notin \mathbb{R}_-^\Phi$.

Similarly, substituting cumulative distribution regret for cumulative action regret yields the generalized Blackwell distribution regret condition.

Definition 15 (Generalized Blackwell Distribution Regret Condition) *Given a reward system $(A, \mathcal{R})$, a finite set $\Phi \subseteq \Phi_{ALL}$ of action transformations, and a link function $f : \mathbb{R}^\Phi \rightarrow \mathbb{R}_+^\Phi$, an online learning algorithm $\{L_t\}_{t=1}^\infty$ is said to satisfy the (Φ, f) -Blackwell distribution-regret condition if*

$$f \left(\hat{R}_{t-1}^\Phi(h) \right) \cdot \mathbb{E}_{a \sim L_t(h)} [\rho^\Phi(a, r)] \leq 0 \quad (37)$$

for all times $t \geq 1$, for all histories $h \in H_{t-1}$ of length $t-1$, and for all reward functions $r : A \rightarrow \mathcal{R}$.

Theorem 16 (Distribution Regret Matching) *Given a real-valued reward system $(A, \mathcal{R})$, a finite set $\Phi \subseteq \Phi_{ALL}$ of action transformations, and a link function $f : \mathbb{R}^\Phi \rightarrow \mathbb{R}_+^\Phi$, the (Φ, f) -distribution-regret-matching algorithm $\{L_t\}_{t=1}^\infty$ satisfies the (Φ, f) -Blackwell distribution-regret condition (with equality).*

The proof is virtually identical to the proof of Theorem ??.

To bound distribution regret, we use a simplified version of Gordon's Theorem.

3. In their original paper, Cesa-Bianchi and Lugosi (2003) failed to account for the non-differentiability of polynomial link functions when applying Taylor's Theorem. Though their original bounds match those obtained here, their corrected bounds are slightly weaker (see <http://homes.dsi.unimi.it/~cesabian/predbook/errata.pdf>).

Theorem 17 Assume $\langle G, g, \gamma \rangle$ is a Gordon triple and $C : \mathbb{N} \rightarrow \mathbb{R}$. Let $X_0 \in \mathbb{R}^n$, let $x_1, x_2, \dots$ be a sequence of random vectors over $\mathbb{R}^n$, and define $X_t = X_{t-1} + x_t$ for all times $t \geq 1$.

If, for all times $t \geq 1$,

$$g(X_{t-1}) \cdot x_t + \gamma(x_t) \leq C(t) \quad a.s. \quad (38)$$

then, for all times $t \geq 0$,

$$G(X_t) \leq G(X_0) + \sum_{\tau=1}^t C(\tau) \quad a.s. \quad (39)$$

Proof The proof is by induction on t . For $t = 0$, the result is immediate. For $t \geq 1$,

$$G(X_t) = G(X_{t-1} + x_t) \quad (40)$$

$$\leq G(X_{t-1}) + g(X_{t-1}) \cdot x_t + \gamma(x_t) \quad (41)$$

$$\leq G(X_{t-1}) + C(t) \quad (42)$$

$$\leq G(X_0) + \sum_{\tau=1}^{t-1} C(\tau) + C(t) \quad (43)$$

$$= G(X_0) + \sum_{\tau=1}^t C(\tau) \quad (44)$$

with all inequalities holding almost surely. ■

We then get the analogous corollary:

Corollary 18 Given a real-valued reward system $(A, \mathcal{R})$ and a finite set $\Phi \subseteq \Phi_{ALL}$ of action transformations. If $\langle G, g, \gamma \rangle$ is a Gordon triple, then a (Φ, f) -distribution-regret-matching algorithm $\{L_t\}_{t=1}^\infty$ guarantees

$$G(\hat{R}_t^\Phi) \leq G(0) + t \sup_{a \in A, r \in \Pi} \gamma(\hat{\rho}^\Phi(a, r)) \quad a.s. \quad (45)$$

at all times $t \geq 1$.

This corollary yields corresponding versions of the specific bounding theorems for polynomial and exponential distribution regret matching. Thus, each bound on the expectation of action Φ -regret that we prove for an (Φ, f) -action-regret-matching algorithm is also a bound on distribution Φ -regret for the analogous (Φ, f) -distribution-regret-matching algorithm.

In summary, for a polynomial (Φ, f) -distribution-regret-matching algorithm with $p > 2$,

$$\max_{\phi \in \Phi} \frac{1}{t} \hat{R}_t^\phi \leq \sqrt{\frac{p-1}{t}} \sqrt[p]{\mu(\Phi)}, \quad (46)$$

and with $1 \leq p \leq 2$,

$$\max_{\phi \in \Phi} \frac{1}{t} \hat{R}_t^\phi \leq t^{(\frac{1}{p}-1)} \sqrt[p]{\mu(\Phi)}; \quad (47)$$

and for an exponential (Φ, f) -distribution-regret-matching algorithm with parameter η ,

$$\max_{\phi \in \Phi} \frac{1}{t} \hat{R}_t^\phi \leq \frac{\ln |\Phi|}{\eta t} + \frac{\eta}{2}. \quad (48)$$

4.4 Summary of Bounds

We have considered two classes of algorithms, polynomial and exponential regret matching, each of which has a single parameter, p and η , respectively. Table 1 summarizes the bounds we derived on $\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right]$ (for action regret-matching) and $\max_{\phi \in \Phi} \frac{1}{t} \hat{R}_t^\phi$ (for distribution regret-matching). Our two analyses of polynomial regret matching (Theorems 9 and 11) agree when $p = 2$. In general, our bounds on polynomial distribution-regret matching for $2 \leq p < \infty$ agree with those of Cesa-Bianchi and Lugosi (2003), although their bounds are computed in terms of the number of experts rather than the number of action transformations (see Section 5 for details). For external and internal polynomial distribution-regret matching, in particular, we improve upon the bounds that can be derived immediately from their results. Though the improvement is small for external regret (from a bound proportional to $\sqrt[p]{|A|}$ to a bound proportional to $\sqrt[p]{|A| - 1}$), it is more significant for internal regret (from $|A|^{2/p}$ to $(|A| - 1)^{1/p}$).

Table 1: Bounds for polynomial and exponential regret-matching algorithms.

$f_i(x)$	Condition	Bound for finite $\Phi \subseteq \Phi_{\text{ALL}}$	Bound for $\Phi_{\text{EXT}}, \Phi_{\text{INT}}$
$(x_i^+)^{p-1}$	$2 < p < \infty$	$\sqrt[p]{\frac{p-1}{t}} \sqrt[p]{\mu(\Phi)}$	$\sqrt[p]{\frac{p-1}{t}} \sqrt[p]{ A - 1}$
$(x_i^+)^{p-1}$	$1 \leq p \leq 2$	$t^{(\frac{1}{p}-1)} \sqrt[p]{\mu(\Phi)}$	$t^{(\frac{1}{p}-1)} \sqrt[p]{ A - 1}$
$e^{\eta x_i}$	$\eta > 0$	$\frac{\ln \Phi }{\eta t} + \frac{\eta}{2}$	$\frac{\ln \Phi }{\eta t} + \frac{\eta}{2}$

Finally, observe that for $p > 1$ polynomial action-regret matching has the property that

$$\lim_{t \rightarrow \infty} \mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq 0 \quad (49)$$

while exponential action-regret matching has the property that

$$\lim_{t \rightarrow \infty} \mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq \frac{\eta}{2} \quad (50)$$

In particular, the bound on the time-averaged action-regret of any polynomial algorithm is eventually better than that of any exponential algorithm. An analogous result holds for distribution regret.

4.5 Optimal Parameters

We now consider the task of optimizing the parameters in the polynomial and exponential regret-matching algorithms so as to minimize the corresponding bounds. For the polynomial algorithms, we derive the optimal value of the parameter p ; previously, only approximately optimal values were reported. Also, we find that no matter how we tune the parameters of an exponential algorithm, any polynomial algorithm eventually achieves a lower bound on its regret. However, for sufficiently large action sets, a properly tuned exponential algorithm obtains a lower bound on its regret at a target time t than any polynomial algorithm.

The bound for polynomial regret matching derived in Theorem 11 and Equation 46 for $1 \leq p \leq 2$ is strictly decreasing in p ; hence $p = 2$ is the optimal setting. By considering the partial derivative with respect to p of the bound derived in Theorem 9 and Equation 47 for $2 < p < \infty$, we find that we can minimize this bound by setting $p = p^*(\Phi)$, defined as

$$p^*(\Phi) = \begin{cases} \psi(\mu(\Phi)) & \text{for } \mu(\Phi) \geq e^2 \\ 2, & \text{otherwise} \end{cases} \quad (51)$$

where $\psi(x) = \ln x + \sqrt{(\ln x)^2 - 2 \ln x}$. Combining these results, we see that in fact $p = p^*(\Phi)$ is optimal for all $1 \leq p < \infty$. This result is similar to the optimal parameter for the p -norm regression algorithm calculated by Gentile (2003). Cesa-Bianchi and Lugosi (2003) suggest setting $p = 2 \ln |\Phi|$, which is nearly optimal when $\mu(\Phi) = |\Phi|$. In particular, if we choose $p = 2 \ln \mu(\Phi)$, although this choice is suboptimal, as Gentile (2003) observes, it yields a simpler bound, which differs from the optimal only by lower order terms.

Considering the partial derivative with respect to the parameter η of the bounds derived in Theorem 13 (action regret) and Equation 48 (distribution regret) for exponential regret matching, we find that the optimal setting of η depends on the time t at which we want to optimize the bound. The best bound at time t^* is obtained by setting $\eta = \eta^*(\Phi, t^*)$, where

$$\eta^*(\Phi, t^*) = \sqrt{\frac{2 \ln |\Phi|}{t^*}}. \quad (52)$$

Plugging $\eta^*(\Phi, t^*)$ into Equation 35, we find that an optimized action-regret-matching algorithm guarantees

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t^*} R_{t^*}^\phi \right] \leq \sqrt{\frac{2 \ln |\Phi|}{t^*}}. \quad (53)$$

The analogous result in the case of distribution regret was obtained by Cesa-Bianchi and Lugosi (2003). For large enough action sets ($|A| \geq 4$ for Φ_{EXT} , $|A| \geq 13$ for Φ_{INT}), an $\eta^*(\Phi, t^*)$ exponential algorithm has a lower bound than any polynomial algorithm at $t = t^*$. For small action sets, however, an optimal polynomial algorithm has a lower bound than any exponential algorithm, for all times t .

5. Related Work

The literature is rife with analyses of regret-minimization algorithms defined within a variety of frameworks. There are at least four dimensions on which these analyses vary. First, as emphasized here, regret may be computed relative to *actions* or *distributions*. Second, different frameworks incorporate different kinds of *transformations* (e.g., Φ_{INT} and Φ_{EXT}), and consequently feature different kinds of regret. Third, there are two broad classes of results about online learning algorithms. *Bounding results*, derived primarily by computer scientists, provide functions that bound the time-averaged (or cumulative) regret vector at a particular time t . *Convergence results*, derived primarily by game theorists, establish guarantees on the behavior of the time-averaged regret vector as $t \rightarrow \infty$. Fourth is the *algorithm* itself. In the case of regret matching, this amounts to choosing, along with the variant of regret, a link function (or equivalently a potential function).

5.1 Regret Varieties

The three most commonly studied forms of regret are external, internal, and swap. The notion of external regret is attributed to Hannan (1957). Indeed, the no-external-regret property is often called “Hannan consistency,” although it is also sometimes called “universal consistency” (Fudenberg and Levine, 1995). Foster and Vohra (1999) introduced the notion of internal regret, and Blum and Mansour (2005) introduced the terminology “swap regret.”

5.2 No Regret

A commonly studied regret property is the “approachability” (in the sense of Blackwell (1956)) of the negative orthant by the vector of time-averaged action regrets. Equivalently (for finite-dimensional regret vectors), one may consider the approachability of $\mathbb{R}_-$ by the maximal entry in the time-averaged action-regret vector. Approachability necessitates uniform convergence: the sequence in question must converge to the set in question at a uniform rate, regardless of the ODP.

Weaker convergence results are also possible. The time-averaged regret vector may converge to a set almost surely or, weaker still, in probability. The notion of “no regret” (e.g., no-external-regret, no-internal-regret) is used by some to indicate approachability of the negative orthant and by others to indicate almost surely convergence to the negative orthant. *We adopt the first interpretation.* Formally, in our Φ -regret framework:

Definition 19 *Given a reward system $(A, \mathcal{R})$ and a finite set of action transformations $\Phi \subseteq \Phi_{ALL}$, an online learning algorithm $\{L_t\}_{t=1}^\infty$ is said to exhibit **no- Φ -regret** if for all $\epsilon > 0$ there exists $t_0 \geq 0$ such that for any ODP, $\Pr \left[\exists t > t_0 \text{ s.t. } \max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \geq \epsilon \right] < \epsilon$.*

By applying the Hoeffding-Azuma lemma (see, for example, the Appendix of Cesa-Bianchi and Lugosi (2006)), it can be shown that a bound on time-averaged distribution regret that converges to zero as $t \rightarrow \infty$ is a sufficient condition for no-regret. Consequently, the polynomial distribution-regret-matching algorithms studied in this paper are no-regret algorithms.

5.3 Convergence Results

The Φ -transformation framework used in this paper was introduced in Greenwald and Jafari (2003) and further studied in Greenwald et al. (2006a). For finite $\Phi \subseteq \Phi_{ALL}$, the latter work shows that Φ -action-regret matching with the polynomial link function and $p = 2$ exhibits no- Φ -regret.

Foster and Vohra (1999) focus their investigations on internal polynomial distribution-regret matching with $p = 2$ and derive an $o(t)$ bound on its cumulative internal distribution regret, which, by the Hoeffding-Azuma lemma, is sufficient for no internal regret.

Hart and Mas-Colell (2001) analyze external and internal action regret, the latter of which they call “conditional regret.” They exhibit a class of no-regret algorithms parameterized by potential functions, which includes the polynomial action-regret-matching algorithm studied here.

Lehrer’s (2003) approach combines “replacing schemes,” which are functions from $\mathcal{H} \times A$ to A , with “activeness functions” from $\mathcal{H} \times A$ to $\{0, 1\}$. Given a replacing scheme g and an activeness function I , Lehrer’s framework compares the agent’s rewards to the rewards that could have been obtained by playing action $g(h_t, a_t)$, but only if $I(h_t, a_t) = 1$, yielding a general form of action regret. Lehrer establishes the existence of (no-regret) algorithms whose action regret with respect to any countable set of pairs of replacing schemes and activeness functions, averaged over the number of times each pair is “active,” approaches the negative orthant.

Fudenberg and Levine (1999) suppose the existence of a countable set of categories Ψ and consider “classification rules,” functions from $\mathcal{H} \times A$ to Ψ . They then compare the agent’s rewards to the rewards that could have been obtained under sequences of actions that are measurable with respect to each classification rule. Their framework, which is a special case of Lehrer’s, also yields action regret. In it, they derive a variant of fictitious play, “categorical smooth fictitious play,” with parameter ϵ that guarantees that the limsup of the maximal entry in the time-averaged action-regret vector converges to $(-\infty, \epsilon]$ a.s., a property they call “ ϵ -universal conditional consistency.”

Young (2004) presents an “incremental” conditional regret-matching algorithm, a variant of internal polynomial action-regret matching with $p = 2$. Rather than playing the fixed point of an $|A| \times |A|$ stochastic matrix, the computation of which is an $O(|A|^3)$ operation, Young incrementally updates the agent’s mixed strategy based on the internal action-regret vector, whose maintenance is only an $O(|A|^2)$ operation. Young argues that his approach yields an algorithm that exhibits no internal regret. Presumably, this result can be generalized to yield an entire class of no-regret algorithms parameterized by action transformations Φ and link functions f .

5.4 Bounding Results

Most bounding results can be found in the computer science learning theory literature. Freund and Schapire (1997) introduce the Hedge algorithm, which in our framework arises as external exponential

Convergence	Action Regret	Distribution Regret
SWAP	Lehrer (2003) Greenwald et al. (2006a) Fudenberg and Levine (1999)	
INT	Hart and Mas-Colell (2001) Young (2004)	Foster and Vohra (1999)
EXT		

Bounding	Action Regret	Distribution Regret
SWAP	<i>the present study</i>	<i>the present study</i> Blum and Mansour (2005)
INT		Cesa-Bianchi and Lugosi (2003)
EXT	Hannan (1957)	Freund and Schapire (1997) Herbster and Warmuth (1998)

Table 2: Related Work organized along three dimensions. Note that SWAP results subsume INT results, which in turn subsume EXT results. However, many of these results are more general than their entry in this table suggests. For example, the framework of Herbster and Warmuth (1998) deals with time-varying experts as well as external regret. Also, an appropriate bound on distribution-regret can imply no-regret: i.e., convergence to zero of action regrets (see the discussion of the Hoeffding-Azuma lemma in Section 5.2).

distribution-regret matching. Inspired by the method of Littlestone and Warmuth (1994), they derive a bound on its external distribution regret.

Herbster and Warmuth (1998) consider a finite set of “experts,” which they define as functions from $\mathbb{N}$ to A . In the context of a prediction problem, they compare the agent’s rewards to the rewards that could have been obtained had the agent played according to each such expert at each time t . They present an algorithm and bound its distribution regret with respect to alternatives constructed by dividing its history into finite-length segments and choosing the best expert for each segment. Bounds on external distribution regret can be obtained by specializing their framework.

Cesa-Bianchi and Lugosi (2003) develop a framework of “generalized” regret. They rely on the same notion of experts as Herbster and Warmuth (1998), but they pair experts $f_1, \dots, f_N$ with activation functions $I_i : A \times \mathbb{N} \rightarrow \{0, 1\}$. At time t , for each i , if $I_i(a_t, t) = 1$, they compare the agent’s rewards to the rewards the agent could have obtained by playing $f_i(t)$. This approach is more general than our action-transformation framework in that alternatives may depend on time. At the same time, it is more limited in that it does not naturally represent swap regret. Their calculations yield bounds on generalized distribution regret.

Blum and Mansour (2005)’s framework is similar to Lehrer’s, but is applied to distribution rather than action regret. Their “modification rules” are the same as Lehrer’s replacing schemes, but instead of activeness functions, they pair modification rules with “time selection functions,” which are functions from $\mathbb{N}$ to the interval $[0, 1]$. The rewards an agent could have obtained under each modification rule are weighted according to how “awake” the rule is, as indicated by the corresponding time selection function. They present a method that, given a collection of algorithms whose external distribution regret is bounded above by $f(t)$ at time t , generates an algorithm whose swap distribution regret (and hence, internal distribution regret) is bounded above by $|A|f(t)$.

5.5 Regret Frameworks: A Comparison

The frameworks of Lehrer (2003), Blum and Mansour (2005), and Fudenberg and Levine (1999), and our action-transformation framework, can all represent external, internal, and swap regret. The frameworks of Cesa-Bianchi and Lugosi (2003), Hart and Mas-Colell (2001), and Foster and Vohra (1999) can represent external and internal regret naturally.

Lehrer’s (2003) framework is very general. In fact, it subsumes the frameworks studied in Cesa-Bianchi and Lugosi (2003), Herbster and Warmuth (1998), and Fudenberg and Levine (1999). However, it also does not allow for partially awake experts as in Blum and Mansour (2005). Nor does it allow for mixed strategies; thus, it does not subsume our action-transformation framework.

6. Conclusion

In this paper, we developed a general framework for analyzing the performance of a general class of online learning algorithms in online decision problems. Using this framework, we derived bounds on the action (and distribution) regret experienced by action (and distribution) (Φ, f) regret-matching algorithms for Φ_{INT} and Φ_{EXT} and for polynomial and exponential link functions. We also calculated the parameter settings that optimize these bounds.

One advantage of our framework is that it may be used to analyze non-smooth link (or potential) functions. Indeed this work was inspired by the observation that Cesa-Bianchi and Lugosi’s (2003) approach, which is based on Taylor’s theorem, cannot be directly applied to algorithms based on polynomial link functions, because they are not twice-differentiable everywhere.

In ongoing work, we are researching algorithms based on alternative link functions, beyond polynomial and exponential. We are also working on generalizing our framework to accommodate activation functions (Lehrer, 2003) and transformations that vary with time.

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A. Proofs of Theorems 9, 11, and 13

We rely on the following lemmas in deriving our bounds.

Lemma 20 *Given a reward system $(A, \mathcal{R})$ and a finite set of action transformations $\Phi \subseteq \Phi_{ALL}$, let f, f' be two link functions, both mapping $\mathbb{R}^\Phi$ to $\mathbb{R}_+^\Phi$. If there exists a strictly positive function $\psi : \mathbb{R}^\Phi \rightarrow \mathbb{R}$ such that $\psi(x)f(x) = f'(x)$ for all $x \in \mathbb{R}^\Phi$, then a (Φ, f) -regret-matching algorithm is also a (Φ, f') -regret-matching algorithm.*

Proof For arbitrary time $t \geq 1$ and for history $h \in H_{t-1}$, let M_t and M'_t be defined according to Equation 11 for f and f' , respectively. Since ψ is strictly positive, $M_t = M'_t$ so that a (Φ, f) -regret-matching algorithm plays a fixed point of M'_t , whenever $R_{t-1}^\Phi(h) \notin \mathbb{R}_-^\Phi$. ■

Lemma 21 *If x is a random vector that takes values in $\mathbb{R}^n$, then $(\mathbb{E}[\max_i x_i])^q \leq \mathbb{E}[\|x^+\|_p^q]$, for all $p, q \geq 1$.*

Proof

$$\left(\mathbb{E} \left[\max_i x_i \right]\right)^q \leq (\mathbb{E} [\|x^+\|_\infty])^q \leq (\mathbb{E} [\|x^+\|_p])^q \leq \mathbb{E} [\|x^+\|_p^q] \quad (54)$$

The first two inequalities follow from the following two facts, respectively: for all $x \in \mathbb{R}^n$,

- $\max_i x_i \leq \max_i x_i^+ = \max_i |x_i^+| = \|x^+\|_\infty$
- $\|x\|_\infty \leq \|x\|_p$, for all $p \geq 1$

The third inequality follows from Jensen's inequality: in particular, $(\mathbb{E} [x])^q \leq \mathbb{E} [x^q]$, for all $q \geq 1$. ■

Lemma 22 *Given a bounded reward system $(A, [0, 1])$ and a finite set of action transformations $\Phi \subseteq \Phi_{ALL}$, $\|\rho^\Phi(a, r)\|_p \leq (\mu(\Phi))^{1/p}$, for all reward functions $r : A \rightarrow [0, 1]$.*

Proof Since rewards are bounded in $[0, 1]$, regrets are bounded in $[-1, 1]$, so that

$$\|\rho^\Phi(a, r)\|_p = \sqrt[p]{\sum_{\phi \in \Phi} (\rho^\phi(a, r))^p} \leq \sqrt[p]{\sum_{\phi \in \Phi} \mathbb{1}_{\phi(a) \neq \delta_a}} \leq \sqrt[p]{\mu(\Phi)} \quad (55)$$

■

Theorem 9 *Given a bounded ODP, a finite set of action transformations $\Phi \subseteq \Phi_{ALL}$, and a polynomial link function f with $p > 2$, a (Φ, f) -regret-matching algorithm guarantees*

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq \sqrt{\frac{p-1}{t}} \sqrt[p]{\mu(\Phi)} \quad (56)$$

at all times $t \geq 1$.

Proof Let $\langle G, g, \gamma \rangle$ be the Gordon triple defined in Lemma 8. By Lemma 20, a (Φ, f) -regret-matching algorithm is also a (Φ, g) -regret-matching algorithm. Now

$$\left(\mathbb{E} \left[\max_{\phi \in \Phi} R_t^\phi \right] \right)^2 \leq \mathbb{E} \left[\left\| (R_t^\Phi)^+ \right\|_p^2 \right] \quad (57)$$

$$= \mathbb{E} [G(R_t^\Phi)] \quad (58)$$

$$\leq G(0) + t \sup_{a \in A, r \in \Pi} \gamma(\rho^\Phi(a, r)) \quad (59)$$

$$= G(0) + t(p-1) \left(\sup_{a \in A, r \in \Pi} \|\rho^\Phi(a, r)\|_p^2 \right) \quad (60)$$

$$\leq t(p-1)(\mu(\Phi))^{2/p} \quad (61)$$

The first inequality follows from Lemma 21, with $x = R_t^\Phi$, $q = 2$, and $2 < p < \infty$. The second inequality is an application of Corollary 7. The third inequality follows from Lemma 22. Hence,

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq \sqrt{\frac{p-1}{t}} \sqrt[p]{\mu(\Phi)} \quad (62)$$

■

Theorem 11 *Given a bounded ODP, a finite set of action transformations $\Phi \subseteq \Phi_{ALL}$, and a polynomial link function f with $1 \leq p \leq 2$, a (Φ, f) -regret-matching algorithm guarantees*

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq t^{(\frac{1}{p}-1)} \sqrt[p]{\mu(\Phi)} \quad (63)$$

at all times $t \geq 1$.

Proof Let $\langle G, g, \gamma \rangle$ be the Gordon triple defined in Lemma 10. By Lemma 20, a (Φ, f) -regret-matching algorithm is also a (Φ, g) -regret-matching algorithm. Now

$$\left(\mathbb{E} \left[\max_{\phi \in \Phi} R_t^\phi \right] \right)^p \leq \mathbb{E} \left[\left\| (R_t^\Phi)^+ \right\|_p^p \right] \quad (64)$$

$$= \mathbb{E} [G(R_t^\Phi)] \quad (65)$$

$$\leq t \sup_{a \in A, r \in \Pi} \gamma(\rho^\Phi(a, r)) \quad (66)$$

$$= t \sup_{a \in A, r \in \Pi} \|\rho^\Phi(a, r)\|_p^p \quad (67)$$

$$\leq t \mu(\Phi) \quad (68)$$

The first inequality follows from Lemma 21, with $x = R_t^\Phi$, $q = p$, and $1 \leq p \leq 2$. The second inequality is an application of Corollary 7. The third inequality follows from Lemma 22. Hence,

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq t^{(\frac{1}{p}-1)} \sqrt[p]{\mu(\Phi)} \quad (69)$$

■

Theorem 13 *Given a bounded ODP, a finite set of action transformations $\Phi \subseteq \Phi_{ALL}$, and an exponential link function f with parameter $\eta > 0$, a (Φ, f) -regret-matching algorithm guarantees*

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq \frac{\ln |\Phi|}{\eta t} + \frac{\eta}{2} \quad (70)$$

at all times $t \geq 1$.

Proof Let $\langle G, g, \gamma \rangle$ be the Gordon triple defined in Lemma 12. By Lemma 20, a (Φ, f) -regret-matching algorithm is also a (Φ, g) -regret-matching algorithm. Now

$$\mathbb{E} \left[\max_{\phi \in \Phi} \eta R_t^\phi \right] \leq \mathbb{E} \left[\ln \sum_{\phi \in \Phi} e^{\eta R_t^\phi} \right] \quad (71)$$

$$\leq \eta \left(G(0) + t \sup_{a \in A, r \in \Pi} \gamma(\rho^\Phi(a, r)) \right) \quad (72)$$

$$= \eta \left(\frac{1}{\eta} \ln |\Phi| + \frac{\eta t}{2} \sup_{a \in A, r \in \Pi} \|\rho^\Phi(a, r)\|_\infty^2 \right) \quad (73)$$

$$= \ln |\Phi| + \frac{\eta^2 t}{2} \quad (74)$$

The second inequality is an application of Corollary 7. Hence,

$$\mathbb{E} \left[\max_{\phi \in \Phi} \frac{1}{t} R_t^\phi \right] \leq \frac{\ln |\Phi|}{\eta t} + \frac{\eta}{2} \quad (75)$$

■